

Reuben Low Yu Xiang

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EDUCATION

Singapore Institute of Technology <i>Bachelor of Engineering in Robotics Systems Engineering (Honours with Distinction)</i>	Sep 2022 – May 2026
Singapore Polytechnic <i>Diploma in Mechatronics & Robotics, and Diploma Plus in Business</i>	Apr 2017 – Mar 2020

COURSEWORK

Robot Operating System (ROS1 & ROS2), Machine Learning, Reinforcement Learning, Robot Dynamics & Control, Computer Vision, Data Structures & Algorithms, Embedded Systems, System Engineering, Project Management

SKILLS

Languages: Python, C, C++
Frameworks & Tools: NVIDIA GR00T (VLA deployment), ROS1, ROS2, MoveIt2, NVIDIA Isaac Sim/Lab, Gazebo, RViz, Docker, Git/GitHub, Linux, MQTT, SolidWorks, Visual Components, LaTeX
Hardware: Unitree H1, Unitree Go2, Kuavo V4/V4 Pro, NVIDIA Jetson AGX Orin, STM32, Arduino, Raspberry Pi, Intel RealSense, Omron TM5-700
Certifications: ROS-Industrial ROS2 Training; ROS-Industrial Consortium Asia Pacific (Sep 2025); Lufthansa-CAS Technical Training (TPE2018-1557)

WORK EXPERIENCE

Ceredroid AI | Robot Learning and Motion Engineer Jan 2025 – Dec 2025

- Developed autonomous manipulation and perception systems on the **Kuavo** humanoid and **Unitree Go2** quadruped, spanning VLM-based task planning and evaluation, visual servoing, and vision-guided hardware payloads (Elaborated under Projects).
- Collected teleoperated demonstrations on the Kuavo humanoid for **NVIDIA GR00T** VLA training, and deployed the trained policies on the robot.
- Developed my **capstone project** in collaboration with the company: a Plan-Act-Correct-Verify (**PACV**) pipeline for long-horizon humanoid manipulation, using VLMs (**Gemini Robotics ER 1.5**, **Cosmos-Reason1**) as planners over **NVIDIA GR00T N1/N1.5**.
- Maintained and upgraded humanoid hardware (Kuavo, Unitree H1): replaced faulty motors, ran firmware updates and recalibration procedures, and authored maintenance SOPs adopted by the team.
- Delivered interactive robotics demonstrations at SIT Open House, RoboSG, and community outreach events.

SIT-FireFense | Robotics Intern Jun 2024 – Aug 2024

- Built an object detection pipeline integrating GPT-4o with an Intel RealSense D435i depth camera, generating 3D pick targets that execute autonomous grasping on an Omron TM5-700 manipulator.
- Developed a Python driver for the Robotiq 2F-85 gripper via MODBUS protocol for the TM5-700.

Singapore Institute of Technology | Student Technical Lab Assistant Jun 2023 – Dec 2024

- Provided technical support for robotics lab operations: soldering, wire crimping, lathe maintenance, and preparing materials and resources for robotics modules.
- Performed equipment repair and maintenance on lab tools and robotic chassis used in coursework.

Singapore Armed Forces | Supply Assistant Jul 2020 – Jul 2022

- Managed logistical operations, maintaining operational readiness and support.
- Awarded Best Soldier of the Month (BSOM), Sergeant Major Coin, and SBW CO Letter of Commendation.

Advanced Remanufacturing and Technology Centre (ARTC) | Engineering Intern Mar 2019 – Aug 2019

- Optimised mirror-finish polishing on stainless steel using the **ABB IRB 140**: experimented with spindle speeds, tool paths, and abrasive grits; analysed and documented surface roughness and visual quality.
- Integrated **ABB's Simplified Robot Programming (SRP)** with the **ABB IRB 6660**; designed 3D-printed spray nozzle adapters and demonstrated motion tracking at the 2019 **ROS-I APAC conference**.
- Developed components for a new **cable management system** on the ABB IRB 6660 robot arm, reducing cable strain and preventing entanglement.

Capstone: Hierarchical Long-Horizon Planning for Humanoid Robotics (Ceredroid AI)*Gemini Robotics ER 1.5, Cosmos-Reason1-7B, NVIDIA GR00T, Python*

- Developed a long-horizon planning pipeline that decomposes natural language goals into subtasks, dispatches them to a **VLA**-driven humanoid, visually verifies each step, and recovers from failures; deployed on the **Kuavo V4 Pro**.
- Iterated from a stateless pipeline to a multi-turn conversation architecture for persistent dialogue context, and constrained VLM-generated plans to the robot's trained skill primitives to prevent hallucinated dispatch.
- Designed a comparative evaluation study across 10,800 scored planning trials with Bonferroni-corrected two-proportion z-tests, testing task decomposition quality, sensitivity to scene complexity, and appropriate refusal of infeasible requests; Gemini ER 1.5 reached **97.2%** success vs **40.8%** for Cosmos-Reason1 on the confirmatory holdout.
- Evaluated long-horizon monitoring on 50 **BEHAVIOR-1K** household tasks in simulation, probing decomposition, progress tracking, and success prediction against simulator ground truth: near-perfect decomposition (6.68/7), moderate progress monitoring (2.01/3), with object hallucination the dominant failure mode.

Storeman-Rosman: Autonomous Dual-Arm Shelf Picking (Ceredroid AI) | Kuavo, ROS, YOLO, Python, REST

- Implemented an autonomous dual-arm pick-and-place system on the Kuavo **humanoid**, with each arm independently fulfilling concurrent customer orders from a shared shelf.
- Defined function-call interfaces between the vision module and motion pipeline, dispatching trajectories from object detections through visual-servoing.
- Collected and annotated a canned-drink dataset across body- and wrist-camera views to fine-tune a **YOLO** detector, and hardened the system with collision-avoidance trajectories and hard-limit safety callbacks for deployment.

Tic-Tac-Toe Humanoid Demo at RoboSG (Ceredroid AI) | Kuavo, ROS, Computer Vision, SolidWorks, Python

- Built an end-to-end demo in which the **Kuavo** played tic-tac-toe against the public, using a three-stage **ROS** pipeline (camera alignment, board classification, piece placement) with state management, command queuing, and blocking logic to prevent race conditions.

Autonomous Bird-Deterrence System (Ceredroid AI) | Unitree Go2, Intel RealSense, YOLO, Arduino, SolidWorks, Python

- Designed and built a vision-guided laser deterrence payload for the **Unitree Go2** quadruped, including the system's electrical schematic, integrating a **YOLO** model with **RealSense** depth sensing.
- Implemented **PID**-based target tracking that converts detections and depth measurements into pan-tilt angles, with multiple laser movement patterns to prevent bird habituation.

Go2 Public Outreach Quadruped (Ceredroid AI) | Android, MQTT, Jetson AGX Orin, SolidWorks, REST

- Developed an Android teleoperation app for **Unitree Go2** integrating **MQTT** communication and the Unitree SportClient SDK; designed mounting hardware for the Jetson AGX Orin, speakers, and portable power to complete the deployable platform.

VLM/LLM Evaluation for Robotics (Ceredroid AI) | NVIDIA Cosmos-Reason1, K2-Think, BitsAndBytes, Python

- Evaluated **Cosmos-Reason1** and **Gemini Robotics ER 1.5** on synthetically generated abnormal-trajectory videos, comparing rule-based versus temporal-description prompting strategies and documenting detection failure cases.
- Evaluated **K2-Think** model via **4-bit quantization**, measuring inference latency and VRAM consumption to establish deployment envelopes.

EASY Pastry Printer | STM32, C, SolidWorks, Pixy Camera, SCRUM, Project Management

- Collaborated with **Nicher**, an artisan bakery, to develop a food printer capable of printing 2D images and logos onto a tray of pastries in any orientation autonomously.

Bolloré Logistics | ROS1, ROS2, UR10, AgileX-LIMO, Visual Components, STM32, C, Python, SolidWorks

- Collaborated with **3PL** company Bolloré Logistics to produce a business case for a modular automated solution that allows for easy integration with emerging technologies for Value Added Service assembly lines.
- Simulated the solution in **Visual Components** and ROS 2 (Gazebo, RViz), then built and integrated a physical **proof-of-concept** of the assembly line (robotic platform and robotic arm).
- Business case projected a **25%** increase in throughput and a corresponding **25%** reduction in costs and labour, with a projected **ROI period of three years**.

Robotic Platforms | C, Python, Spin Language, STM32, Pixy Camera, Parallax MCU, AgileX-LIMO

- Developed a robotic platform capable of remote control via a GUI, further evolving it into an **autonomous system** using a Pixy camera to **follow coloured objects**.
- Implemented Breadth-First Search Algorithm in **ROS 2** Gazebo and RViz to explore an unknown maze.
- Developed scripts for autonomous goal-to-goal navigation, incorporating MoveBaseAction and MoveBaseGoal packages, implemented a GUI with **PyQt**, functionalities to save and load robot positions. Optimised navigation behaviour via ROS parameter YAML files.